

David Xia

davidxia143@gmail.com |  LinkedIn - dx3 |  GitHub - davidxia3

EDUCATION

• University of Illinois Urbana-Champaign

August 2023 - May 2027

Bachelor of Science in Computer Science

Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration

Bachelor of Science in Liberal Arts & Sciences Major in Statistics

- GPA: 3.98/4.00

Champaign, Illinois

EXPERIENCE

• Carnegie Mellon University Human-Computer Interaction , Research Intern

May 2025 - Present

- Mentored by Prof. Jeff Bigham and Prof. Chris Maury to promote local civic engagement using LLMs for city council meeting reporting.
- Evaluating and developed an end-to-end LLM pipeline for identifying newsworthy topics and generating engaging headlines.
- Presented research poster at the CMU HCII REU Symposium and currently writing a paper.

• Illinois Combinatorics Lab for Undergraduate Experience (ICLUE) , Researcher

Jan 2025 - Present

- Mentored by Prof. Alexander Yong and Ph.D. student Duy Phan, studying algebraic combinatorics.
- Co-authored a paper on and currently researching the Robinson-Schensted-Knuth correspondence.

• National University of Singapore SEEDER Group , Research Intern

May 2024 - July 2024

- Mentored by Prof. Kelvin Xuanyao's SEEDER Group to developed GARDEN, a machine learning pipeline to detect, classify, and analyze plant diseases, improving previous accuracies by 40%.
- Implemented generative, classification, and segmentation models built utilizing PyTorch and TensorFlow framework, using self-collected data from local aquaculture farms, and presented findings in manuscript.

• University of California San Diego, Research Intern

June 2020 - August 2023

- Mentored by Prof. Xinyu Zhang and on various IoT and mobile device projects over four years.
- Collaborated with Prof. Ke Sun to develop EveGuard, a software framework that defends against smartphone side-channel eavesdropping using an adversarial perturbation generator model.
- Trained segmentation and classification models for CogTeach, a facial expression recognition application tool for virtual learning, and conducted pilot study by delivering courses of lectures on neural networks.

PROJECTS

• LLMs for Extracting Newsworthiness and Engaging Headlines, Python

May 2025 - Present

- Goal of promoting local civic engagement by using LLMs for affordable city council meeting reporting.
- Evaluating and developed an end-to-end LLM pipeline for identifying newsworthy topics and generating engaging headlines.
- LLM generated headlines consistently drew stronger interest and opinions from crowd sourced human-evaluators on Prolific than expert written headlines.
- The LLMs also prioritized more interesting and newsworthy topics compared to expert selected topics, showing that LLMs can perform the modular task of covering city council meetings better than the expert.
- Presented research poster at the CMU HCII REU Symposium and currently writing a paper.

• EveGuard: Software framework for eavesdropping defense, TensorFlow, GAN May 2022 - August 2023

- Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality.
 - Implemented Eve-GAN, a novel domain translation task for inferring eavesdropped signals, enabling end-to-end training of PGM, utilizing few-shot learning techniques to reduce data collection overhead.
 - Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
 - Co-authored a paper due to appear in the IEEE Symposium on Security and Privacy 2025.
- **GARDEN: ML pipeline for anomaly detection**, PyTorch, TensorFlow [🔗] May 2024 - July 2024
 - Machine learning pipeline project at the NUS SEEDER Group.
 - Developed machine learning pipeline to detect, classify, and analyze diseases found in plant leaves.
 - Trained U-Net segmentation models and classification models based on convolutional neural networks to isolate and classify images using the TensorFlow framework, achieving 90% segmented classification rate.
 - Generated synthetic images using custom trained cGAN model to increase data variety and volume.
 - Stacked deep learning models with traditional techniques like support vector machines, gradient boosting, logistic regression, and decision trees to achieve an increased detection accuracy of over 95%.
 - **Predictability and Competitiveness of Sports Leagues**, Pandas [🔗] [🔗] January 2024 - Present
 - Researched the data science perspective of sports competitions with Prof. AJ Hildebrand at the Illinois Mathematics Lab.
 - Built web scrapers with Selenium to scrape online data and analyzed the predictability of sports matches by comparing prediction accuracies of multiple sources including polls and the betting market.
 - Evaluated efficiency of probabilistic predictions based on money line, Elo, and other models.
 - Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, and Joint Mathematics Meeting 2025.

PRESENTATIONS

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| 2025 | Carnegie Mellon University Human-Computer Interaction Institute REU Symposium LLMs for Extracting Agenda Items from City Council Meetings and Generating Newsworthy Headlines |
| 2025 | Algebra, Geometry and Combinatorics Day RSK linear operators and the Vershik-Kerov-Logan-Shepp curve (presented by Duy Phan) |
| 2025 | Joint Mathematics Meeting Pi Mu Epsilon Poster Session Predictability in Major Sports Leagues |
| 2024 | Illinois Mathematics Lab Research Symposium Predictability in College Sports |
| 2024 | University of Illinois Undergraduate Research Symposium Predictability in College Sports |
| 2024 | Rose-Hulman Undergraduate Math Conference Predictability in College Sports |

PAPERS

PRE=PREPRINT, ACC=ACCEPTED

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| [1] | [PRE] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve [arXiv]
Duy Phan, David Xia
<i>Preprint</i> |
| [2] | [ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations [arXiv]
Jung-Woo Chang, Ke Sun, David Xia, Xinyu Zhang, Farinaz Koushanfar
<i>IEEE Symposium on Security and Privacy, 2025</i> |